

PULKIT BANSAL

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TECHNICAL SKILLS

Frameworks: Spring Boot · Angular

Languages: Java · Python · C++ · TypeScript · JavaScript · SQL · React

Backend & APIs: Spring Boot · RESTful · microservices · OOP design patterns · Mockito · Selenium

DevOps & Tooling: Docker · GitHub Actions · CI/CD · Git · Linux CLI

Database & Pipelines: PostgreSQL · Oracle · Hibernate

PROJECTS

ClaimsAI — AI-Powered Insurance Claims Management Platform(Java · Spring Boot · Angular · PostgreSQL · Docker)

<https://pulkit2306.github.io/ClaimsAI/>

- Architected 6 Spring Boot microservices (Auth, Policy, Claims, AI, Search, API Gateway) with event-driven communication via Apache Kafka and JWT/OAuth2 security.
- Built RAG pipeline using Spring AI + pgvector for AI-powered claims chat, fraud detection, and automated claim summarization.
- Implemented semantic search with Elasticsearch multi-match queries, fuzzy matching, and MongoDB audit trail for compliance.
- Containerized full stack (14 containers) with Docker Compose; one-command deployment with health-check dependencies and multi-stage builds.

QuantFrame — Algorithmic Trading Data Pipeline & Backtesting Engine(C++ · Python · PostgreSQL · Docker · Linux)

<https://pulkit2306.github.io/QuantFrame/>

- Developing a C++ order book engine with concurrent, multi-threaded data structures and nanosecond-precision timestamps for backtesting trading strategies.
 - Built an async market-data ingestion pipeline in Python (10+ symbols, OHLCV/NBBO quotes) with concurrent batch fetching and rate-limit handling.
 - Designed a partitioned PostgreSQL schema (BRIN indexes, composite keys), with an idempotent recovery process for safe restart after failure.
 - Built a full automated test suite (pytest) with continuous integration on Linux via GitHub Actions, plus a CLI for monitoring pipeline behavior.
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WORK EXPERIENCE

Full Stack Developer Intern · Tech Mahindra COE, India

Jan 2024 – Jun 2024

- Shipped RESTful APIs (Spring Boot/Python microservices) improving throughput by 20%.
 - Applied OOP design patterns (strategy, repository, factory) across large Java codebase using hibernate for ORM-based data persistence.
 - Automated CI/CD with GitHub Actions + Docker, cutting deployment time by 40%
 - Led code reviews resolving 30+ critical issues in Agile sprints.
 - Implemented test suite on different features using tools like Selenium, Mockito and created database schemas using Oracle.
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EDUCATION

M.Eng. Software Engineering

Concordia University, Montréal, QC · Sept 2024 – June 2026

Core Exec, Graduate Students Association (2025–26)

B.Sc. Computer Science and Engineering

Chitkara University, India · Oct 2020 – Aug 2024

Core Exec, ACM Student Chapter (2022–23)